

Energy Storage State-of-Charge Management in Real-Time Markets

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Abstract—The evolving resource mix will likely increase net load variability and forecast errors. While numerous studies show that grid-scale energy storage resources (ESRs) can mitigate reliability issues, they often assume perfect foresight of net load when optimizing their State-of-Charge (SoC). In practice, electricity markets operate on a rolling horizon, frequently updating short-term load forecasts. At each market clearing, operators must allocate SoC between immediate and later needs under forecast uncertainty. Through simulations of operator-managed ESRs within PJM’s real-time markets under a future resource mix, we demonstrate that both a greedy strategy that prioritizes smoothing load volatility as it occurs and a conservative strategy that holds energy for peak load arbitrage can exacerbate reliability issues. We propose an SoC management framework that balances between the greedy and conservative strategies, improving reliability by absorbing near-term net load shocks while maintaining SoC for expected peak load beyond the horizon of real-time processes.

Index Terms—Energy storage, net load variability, forecast uncertainty, SoC, markets

I. INTRODUCTION

AS the power system’s resource mix transitions from being dominated by fossil-fuel-fired thermal generators to one that is increasingly reliant on renewable energy sources (e.g., wind, solar) and energy-limited resources (e.g., energy storage), net load variability and forecast error (intra-day and intra-hour) are likely to increase. Many anticipate that grid-scale energy storage resources (ESRs) will provide cost-effective management of net load variability to maintain reliability. ESRs can provide system benefits as wholesale market participants, but their effectiveness will depend critically on how the State of Charge (SoC) is managed.

Numerous studies have focused on optimal bidding strategies for ESRs from the resource owner’s perspective (e.g., [1–4]). These studies take energy and ancillary service prices as given and use them to develop optimal bidding strategies for an ESR. However, these studies do not explicitly model how the system operator would alter the commitment and dispatch of generators in response to the presence of the ESRs.

System-level studies take the ISO/RTO perspective and explore how the addition of ESRs would impact the commitment and dispatch of all system resources and affect prices. These studies generally assume that the operator can control the aggregate SoC of the fleet of ESRs, and focus on the question of how the SoC should be managed within an operating day. Several studies present stochastic or robust unit commitment approaches for the hourly day-ahead problem (e.g., [5–8]).

Other studies have considered operating reserve eligibility for ESRs [9], price impacts from alternative multi-settlement designs [10], or unified commitment and dispatch frameworks [11]. Goyal et al. [8] modeled ESRs as participants in a two-settlement market, with a day-ahead model setting an hourly SoC schedule, then studied individual hours at the 5-minute scale in Real-Time (RT) markets. However, their model did not consider the accumulation of errors over an operating day, which has important implications for managing an energy-limited resource.

Most system-level ESR studies to date optimize SoC over the model horizon (e.g., one day) assuming that net load is known in advance, or, in some cases, simulate a second stage with a single realization for the real-time net load, which is then re-optimized within a shorter horizon. These simulated markets contrast with how real-time markets unfold over each operating day. In North American electricity markets, a day-ahead (DA) market determines the commitment schedule for the next day. Throughout the operating day, operators solve a rolling horizon real-time unit commitment (RTUC) and a real-time economic dispatch (RTED), with updated short-term forecasts before each execution of RTUC and RTED.

ESRs are highly flexible resources, but their energy-limited nature makes SoC management challenging, especially as market operators must make near-term decisions without complete information on how the day will unfold. The main challenge is to leverage the flexibility of ESRs to manage net load volatility as it occurs, while preserving some of the energy for needs beyond the horizon of real-time processes.

In this paper, we present a framework for an RTO/ISO to manage the aggregate SoC over a day within a rolling horizon real-time energy market. We apply a simulation of PJM’s energy markets for a future resource mix including ESRs to compare alternative strategies for managing SoC under uncertainty. We demonstrate for an example day that a greedy strategy that utilizes ESRs to address unexpected net load shocks as they occur can lead to worse reliability later in the day. Conversely, an overly conservative approach that maintains SoC for expected (peak) load shocks in the future may also cause reliability issues. We propose a balanced approach that addresses both near-term net load deviations and prepares for expected future needs. This approach consists of using the downscaled day-ahead SoC schedule in real-time operations, but allowing SoC target violations with an

appropriate penalty. The proposed method provides improved reliability compared with greedy or conservative approaches and is easily implementable within existing real-time market clearing engines.

The remainder of the paper is presented as follows. Section II presents the methodology. Section III presents the results. Section IV provides the conclusion and outlines future work.

II. METHODOLOGY

We apply a real-time market simulation framework that reproduces key features of wholesale energy markets. The model simulates rolling horizon real-time commitment and dispatch with net load forecasts and residual forecast errors changing throughout each day. For each day simulated, the DA is solved to determine the commitment schedules for slow-start units. The real-time operating day is then solved iteratively, with a RTUC to obtain fast-start units' commitment states for the next 15 minutes, followed by a sequence of three single-interval RTED clearings. The model then advances 15 minutes, the net load forecast is updated, and the RTUC is solved for this new horizon, initializing the system state from the previous RTED solution. A new short-term load forecast is obtained before each RTUC and RTED execution. After each dispatch, we solve a real-time pricing run (RTPR), which relaxes integrality constraints to simulate fast-start pricing. Fig. 1 shows a schematic representation of the temporal and information sequence of our simulation framework.

In our model, we represent ESR units in the DA and RT markets. The hourly SoC schedule is obtained from the DA solution, based on the day-ahead load forecast. SoC management in the real-time market consists of three components: i) every execution of the RTUC re-optimizes SoC within its time horizon in 15-min. intervals, using the starting system state, the DA SoC target at the end of RTUC's horizon (e.g., 2 hours ahead), and the updated net load forecast; ii) each single-interval RTED must meet or exceed an SoC target, which is interpolated from the RTUC's target to the 5-minute scale; and iii) the RTED can violate the SoC constraint at a fixed penalty. The penalty price for SoC violations then provides a design parameter that determines the amount of SoC schedule violations that will occur over the day.

A. Model Formulation

We present the changes to the mathematical formulation required to incorporate ESRs and the proposed SoC management scheme into the rolling horizon market simulation framework. Here, we present only the formulation changes related to ESRs. The complete formulations for the DA, RTUC, RTED, and RTPR are provided in [12].

1) *Day-Ahead (DA) Model*: The DA model, formulated as an MILP, solves the commitment and dispatch for a 24-hour horizon in hourly time steps, based on the day-ahead net load forecast. The DA solution includes the optimal SoC schedule over the 24 hours, which provides information about expected needs later in the day for the real-time clearing engines that have shorter horizons.

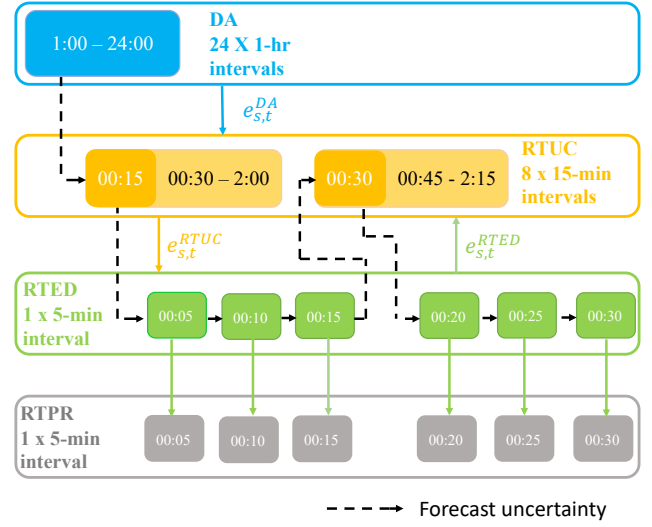


Fig. 1: Market Simulation Framework

The index $g \in G$ represents generators, $t \in T$ denotes time periods, $s \in S$ represents individual ESR units, and $j \in J$ indicates operating reserve demand blocks. The binary variable $u_{g,t}$ indicates the commitment status of generator g at time t , while $u_{s,t}^{ch}$ and $u_{s,t}^{dis}$ indicate charging and discharging status, respectively, for storage s at time t .

$$\min \sum_{g,t} c_{g,t} + \sum_{s,t} C_s^{vom} (p_{s,t}^{dis} + p_{s,t}^{ch}) + \sum_t C_t^{rul} s_t^{ul} + \sum_{j,t} C_j^{sr} s_{j,t}^{sr} \quad (1)$$

$$\sum_g p_{g,t} + \sum_s p_{s,t}^{dis} - \sum_s p_{s,t}^{ch} + s_t^{ul} \geq D_t \quad \forall t \quad (2)$$

$$\sum_g r_{g,j,t}^{sr} + \sum_s p_{s,j,t}^{sr} + s_{g,j,t}^{sr} \geq \sum_j R_j^{sr} \quad \forall t \quad (3)$$

$$e_{s,t} = e_{s,t-1} + \eta^{ch} p_{s,t}^{ch} \tau - \frac{p_{s,t}^{dis}}{\eta^{dis}} \tau \quad \forall s, t \quad (4)$$

$$\underline{E}_s \leq e_{s,t} \leq \overline{E}_s \quad \forall s, t \quad (5)$$

$$\underline{P}_s u_{s,t}^{ch} \leq p_{s,t}^{ch} \leq \overline{P}_s u_{s,t}^{ch} \quad \forall s, t \quad (6)$$

$$\underline{P}_s u_{s,t}^{dis} \leq p_{s,t}^{dis} \leq \overline{P}_s u_{s,t}^{dis} \quad \forall s, t \quad (7)$$

$$u_{s,t}^{ch} + u_{s,t}^{dis} \leq 1 \quad \forall s, t \quad (8)$$

$$p_{s,t}^{sr} \tau \leq (e_{s,t} - \underline{E}_s) \eta^{dis} \quad \forall s, t \quad (9)$$

$$p_{s,t}^{sr} + p_{s,t}^{dis} - p_{s,t}^{ch} \leq \overline{P}_s \quad \forall s, t \quad (10)$$

$$e_{s,t} = T \overline{C} \overline{E}_s \quad \forall s, t \in \{t_0, t_{|T|}\} \quad (11)$$

The objective function (1) is modified to include the costs of charging and discharging. C_s^{vom} represents variable O&M costs of ESRs, and the variables $p_{s,t}^{ch}$ and $p_{s,t}^{dis}$ are the charging and discharging rate of energy storage s at time t . s_t^{ul} and C_t^{rul} denote the amount of unserved load in the system and its associated penalty parameter respectively. Similar, $s_{g,j,t}^{sr}$ and

C_j^{sr} are synchronized reserve shortage amount and penalty. $c_{g,t}$ denotes the total cost of generation including startup and shutdown costs, for each generator g at time t .

The demand (2) and reserve balance constraints (3) are modified to include the effects of charging, discharging, and synchronized reserve allocation $p_{s,t}^{sr}$ from ESR s at time t . The formulation for primary and operating reserve are identical to (3), and are not included here. The remaining constraints (4) - (11) are new constraints added to the DA to represent the physical constraints associated with ESRs. (4) calculates the SoC for each ESR at time t , as a function of the previous interval's SoC and the amount of energy charged or discharged, and τ converts the instantaneous power to energy based on the model's time resolution. Energy losses occur during charging and discharging and are parameterized with the efficiency parameters, η . Minimum and maximum energy levels ($\underline{E}_s$ and $\overline{E}_s$) are enforced by eq. (5) to minimize battery degradation from deep cycles. Equation (6) constrains the maximum instantaneous power flow $\overline{P}_s$ during charging, while (7) limits the maximum instantaneous power flow during discharging. The logical constraint (8) requires that each ESR unit is charging, discharging, or idle in each interval. Constraint (9) ensures that ESRs are able to discharge continuously for at least $\mathcal{T}$ periods to qualify to provide synchronized reserves. Constraint (10) ensures that the total discharging/charging at interval t and any contribution to synchronized reserve is within its maximum power rating. Equation (11) sets the initial and terminal SoC to TC percent of each unit's maximum energy level. The index t_0 is the initial time period and $t_{|T|}$ is the final period, where $|T|$ is the cardinality of set T .

2) *Real-Time Unit Commitment (RTUC) Model:* The RTUC model, formulated as an MILP, solves the unit commitment problem for a two-hour horizon of 15-minute intervals (8 periods) to determine startup decisions for fast-start generators. In this study, the eligible fast-start units are combustion turbine (natural gas and oil) and internal combustion (natural gas and diesel) units. The commitment and startup solutions from only the first 15-minute interval are binding.

To represent ESRs in RTUC, all of the modifications and new constraints (1)-(10) in the DA are also added to RTUC. The following constraints replace (11):

$$e_{s,t} = e_{s,t}^{RTED} \quad \forall s, t = t_0 \quad (12)$$

$$\sum_s e_{s,t} + s^{ue} \geq \sum_s e_{s,t}^{DA} \quad \forall t = t_{|T|} \quad (13)$$

The initial state of charge is set to the actual energy level of ESR from the immediately preceding RTED solution (12). Similarly, in (13), we set the terminal SoC (8th period) based on the corresponding DA solution for SoC, after interpolating from hourly to 15-minute resolution. We introduce a slack variable, s^{ue} , to allow violations when necessary to address reliability issues. The objective function (1) is modified to include the cost of SoC constraint violations ($C^{ue}s^{ue}$).

3) *Real-Time Economic Dispatch (RTED) Model:* The RTED, formulated as a linear programming problem, co-optimizes energy and reserves for a single 5-minute interval.

The formulation is the same as the DA and RTUC except that the commitment status is fixed to the DA and RTUC solutions, and are treated as parameters.

The changes to the DA and RTUC described above are also made to the RTED. In addition, the following constraints replace (12)-(13).

$$e_{s,t} = e_{s,t}^{RTED} \quad \forall s, t = t_0 \quad (14)$$

$$\sum_s e_{s,t} + s^{ue} \geq \sum_s e_{s,t}^{RTUC} \quad \forall t = t_{|T|} \quad (15)$$

Constraint (14) ensures that the initial state of charge is the actual energy level of ESR from the immediately preceding RTED solution. There is no incentive for ESR to charge and maintain energy level in the absence of a constraint because within a myopic single-interval dispatch, discharging the maximum amount possible will always reduce costs. To address this, we introduce constraint (15) to guarantee that the SoC of ESR s meets or exceeds the energy level scheduled by RTUC. For example, if RTUC solves for the SoC of unit s in each interval between 1:00 pm to 3:00 pm, in 15-minute intervals, RTED interpolates the 1:00 to 1:15 SoC value to 5-minute values to obtain the terminal SoC requirement for the next three RTED solutions. The slack term s^{ue} in (15) is added so that the SoC constraint can be violated, with a corresponding penalty in the objective function, if necessary to avoid reliability issues. The objective function (1) is modified to include the cost of SoC target violations ($C^{ue}s^{ue}$) as in RTUC. Allowing SoC constraint violations in RTED with a fixed penalty is critical for balancing short-term needs and terminal conditions when the net load diverges from the forecast.

After each RTED, the RTPR model is solved to obtain real-time energy and reserve prices. The commitment states from RTUC for eligible online units are relaxed to be continuous between 0 and 1. This enables the startup cost of these units to be reflected in the marginal price. All modifications to the RTED to incorporate ESRs are also made to RTPR.

B. Capacity Mix, Forecast Errors, and Model Cases

We apply the model to a case study of PJM's real-time markets. PJM's historical 2023 demand and renewable generation profiles [13] are each scaled by projected growth and then combined to derive net load. We simulate forecast error with an ARIMA model fitted to observed errors in 5-min. load, solar, and wind forecasts in PJM from 2015-2017, also scaled by the projected growth in the future scenario. The error model creates net load profiles that diverge every five minutes to simulate updated forecasts throughout the day. Generator outages are randomly sampled to replicate PJM's historical outage rates.

We assume a future resource mix with a 40% reduction in coal capacity, 8% reduction in natural gas, 500% increase in wind, and 300% increase in solar capacity, relative to PJM's 2023 capacity mix [13]. Generator cost parameters are obtained from S & P Global's Market Intelligence Platform SNL Energy dataset [14]. Gross load is reduced by generation from solar, wind, hydro, and nuclear to obtain net load.

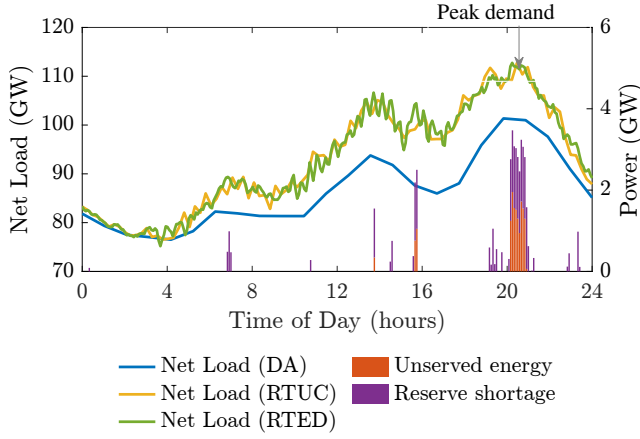


Fig. 2: Net load profile (left y-axis) for example day in DA, RTUC, and RTED with unserved energy and reserve shortage in the Base case (right y-axis)

We assume 40 ESR units with a combined 12 GW of power and 56 GWh of energy. Eight ESRs have 500 MW power and 3000 MWh capacity (6-hour duration), and the remaining 32 have 250 MW power and 1000 MWh (4-hour duration) capacity. All units have a 0.96 efficiency rate, and operate between 20% and 80% of capacity. We assume variable O&M costs of \$3/MWh.

We present five versions of our model for one example day. Base is the reference case with no ESRs. Three versions use a rolling horizon formulation that includes ESRs (Greedy, Balanced, and Conservative), which vary in terms of the penalty for SoC target violations in RTED: \$50/MWh for Greedy, \$220/MWh for Balanced, and \$11,000/MWh for Conservative. The Full Horizon version intertemporally optimizes over the entire day at 5-min. resolution with perfect foresight of the realized net load. All models use the same DA solution except for Base, which solves the DA without ESRs. All models solve for identical realized net load, and all except Full Horizon receive the same short-term forecasts at each execution of RTUC and RTED. The sample day is representative of many of the 365 net load profiles simulated. Simulations were conducted on a Linux cluster with 2.8 GHz 20-core Intel Xeon processors, 256 GB RAM, using GAMS 39.1.0 with CPLEX 12.10.0 and MATLAB for data management and control flow.

III. RESULTS

Fig. 2 shows the net load profile in the DA, RTUC, and RTED models. For this example day, the actual load realized in RTED is higher than the DA forecast. The load shown in Fig. 2 represents the final realized load after the day has concluded. Peak net load is observed at hour 20, with additional, smaller peaks occurring at hours 6 and 14. We indicate the intervals with unserved energy and synchronized reserve shortages from the Base (no ESRs) version.

We present the impact of ESRs on system reliability under three SoC management strategies within a rolling horizon market: Greedy, Conservative, and Balanced (Fig. 3). We

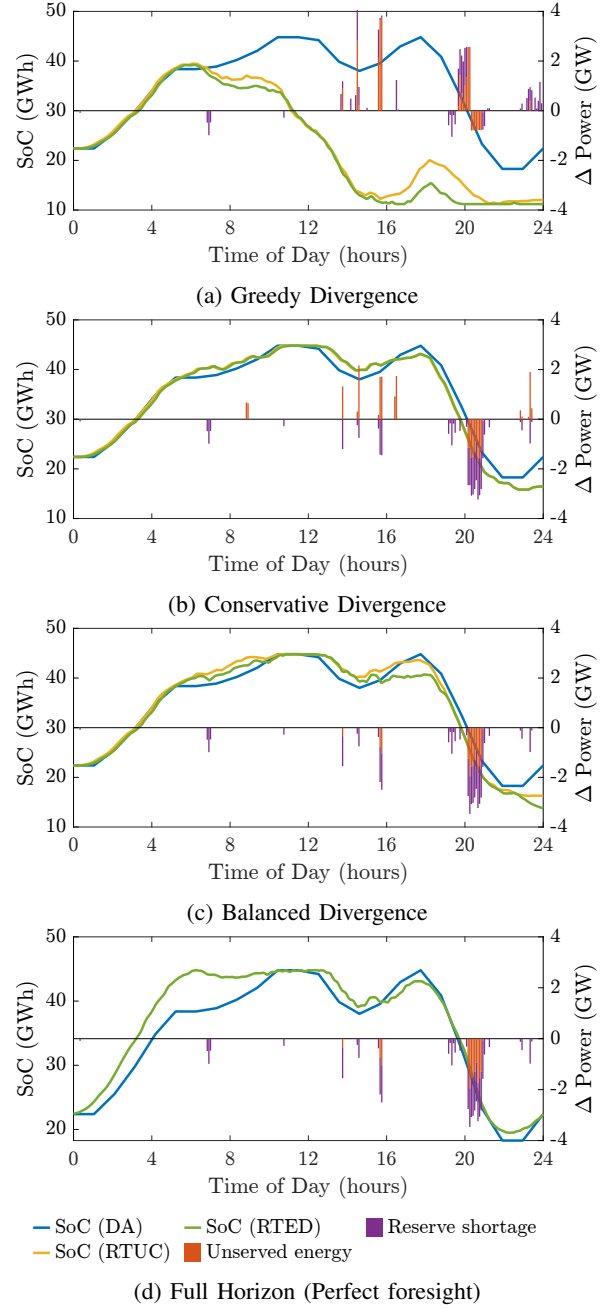


Fig. 3: SoC profile (left y-axis) with unserved energy and reserve shortage (right y-axis) relative to the Base case; all figures use the legend in 3d

compare these results to the Full Horizon version with perfect foresight of the real-time net load over the entire day.

Under the Greedy strategy (Fig. 3a), SoC constraint violations occur whenever the \$50/MWh penalty is less than the cost of meeting the SoC targets. With low penalties, ESRs are discharged aggressively in the first half of the day. RTUC aims to meet the DA SoC target by the end of its horizon, but starts from the previous system state and uses an updated net load forecast. Therefore, the SoC targets from both RTUC (yellow

line) and the actual SoC after each RTED (green line) diverge sharply from the original DA schedule (blue line). Because ESRs are depleted earlier in the day, there are unserved energy and reserve shortages at several intervals that exceed the results with no energy storage.

The Conservative strategy (Fig. 3b) assumes a high penalty of \$11,000/MWh. As a result, there is minimal deviation from the DA SoC schedule, despite the fact that the realized net load diverges from the DA load forecast. Under this strategy, SoC is held during the middle of the day to be able to discharge later during peak hours. This approach reduces unserved energy that occurred in the reference case during the peak hours, but does not eliminate all unserved energy by not responding to the short-term net load volatility.

The Balanced strategy is achieved with a penalty of \$220/MWh, which eliminates unserved energy and reduces contingency reserve shortages (Fig. 3c). With a moderate penalty for SoC violations, ESRs can address both short-term volatility and system needs during later peak periods.

We contrast the rolling horizon results with those from the Full Horizon version. The Full Horizon version represents the approach used in many studies on energy storage. These studies (e.g., [8–11, 15]) focus on load leveling and energy arbitrage by solving a single optimization over the entire day. Studies that do represent uncertainty in the day-ahead typically have a single realized net load profile in real-time that differs from the DA, but is known to the RT engine over the entire day. The Full Horizon version (Fig. 3d) optimally schedules the SoC to be able to eliminate all unserved energy and reserve shortages. The Full Horizon assumes knowledge of the realized net load over the day. In reality, grid operators do not have access to the true net load profile to the end of the day, and must operate within a rolling horizon with imperfect information.

Table I summarizes the aggregate unserved energy and reserve shortage over the sample day. The Balanced strategy eliminated the reliability issues observed in the Base and the other two rolling horizon cases. The reliability outcomes of Balanced case with load forecast uncertainty are identical to the Full Horizon with perfect foresight of load.

TABLE I
RELIABILITY OUTCOMES FOR THE SAMPLE DAY

GWh	Base	Greedy	Conserv.	Bal.	Full Hor.
Unserved Energy	1.27	2.82	1.45	0	0
Reserve Shortage	2.41	3.64	0	0	0

IV. CONCLUSION

This article proposes a methodology for ISO/RTOs to manage SoC under short-term forecast errors and net load volatility. Our approach balances divergence from the DA SoC schedule to address unexpected reliability issues in the short-term with maintaining sufficient energy for expected future needs.

We propose a Balanced strategy by selecting a moderate penalty for SoC violations, which reduces reliability issues

in the system. Further research is needed to develop a robust market design that would optimize the penalty choice for a wider range of future conditions.

The benefits of ESR may be enhanced by real-time market changes, such as flexible ramp products, multi-interval dispatch, or stochastic clearing. Future studies should explore the interaction between market design changes and the utilization of flexible resources such as ESRs in real-time markets with uncertain short-term forecasts.

ACKNOWLEDGMENTS

This work was funded in part by the U.S. Dept. of Energy, ARPA-E # DE-AR0001281 and PJM Interconnection. Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer.

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